Python Lab Assignment
NumPy Practice Questions

1. Create a 3x3 matrix with values ranging from 0 to 8.
2. Create a 2d array with 1 on the border and 0 inside.
3. Replace all odd numbers in the given array with -1.
4. Convert a 1-D array into a 2-D array with 3 rows.
5. Write a Python program that multiplies two 3x3 matrices using numpy's dot() function.
6. Create a numpy array with numbers from 1 to 20 and slice the array to print:
    - The first 5 elements
    - The last 5 elements
    - Elements at even indices
7. Create two numpy arrays a = np.array([1, 2, 3]) and b = np.array([4, 5, 6]). Perform element-wise addition, subtraction, multiplication, and division of these arrays and print the results.
8. Given a numpy array data = np.array([10, 15, 7, 22, 17]), calculate and print the following statistics:
    - Mean
    - Standard deviation
    - Median
    - Sum of all elements

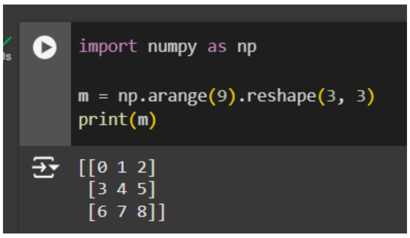
#ANSWERS

#Answer 1

import numpy as np

m = np.arange(9).reshape(3, 3)

print(m)



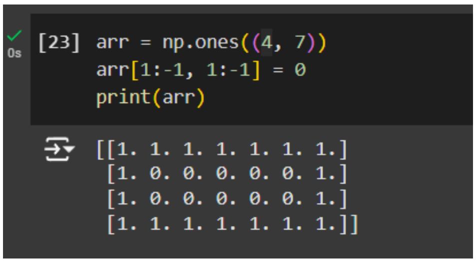
```
import numpy as np

m = np.arange(9).reshape(3, 3)
print(m)

[[0 1 2]
 [3 4 5]
 [6 7 8]]
```

#Answer 2

arr = np.ones((4, 7))

arr[1:-1, 1:-1] = 0

print(arr)



```
[23] arr = np.ones((4, 7))
     arr[1:-1, 1:-1] = 0
     print(arr)

[[1. 1. 1. 1. 1. 1. 1.]
 [1. 0. 0. 0. 0. 0. 1.]
 [1. 0. 0. 0. 0. 0. 1.]
 [1. 1. 1. 1. 1. 1. 1.]]
```

#Answer 3

arr = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9])
arr[arr % 2 == 1] = -1
print(arr)

```
[18] arr = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9])
     arr[arr % 2 == 1] = -1
     print(arr)

[-1  2 -1  4 -1  6 -1  8 -1]
```

#Answer 4

oneD_array = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9])
twoD_array = oneD_array.reshape(3, -1)
print(twoD_array)

```
[24] oneD_array = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9])
     twoD_array = oneD_array.reshape(3, -1)
     print(twoD_array)

[[1 2 3]
 [4 5 6]
 [7 8 9]]
```

#Answer 5

mat_1 = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])
mat_2 = np.array([[9, 8, 7], [6, 5, 4], [3, 2, 1]])

result = np.dot(mat_1, mat_2)
print(result)

```
[20] mat_1 = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])
     mat_2 = np.array([[9, 8, 7], [6, 5, 4], [3, 2, 1]])

     result = np.dot(mat_1, mat_2)
     print(result)

[[ 30  24  18]
 [ 84  69  54]
 [138 114  90]]
```

#Answer 6

array = np.arange(1, 21)

# The first 5 elements
print(array[:5])

# The last 5 elements
print(array[-5:])

# Elements at even indices
print(array[1::2])

```
[11] array = np.arange(1, 21)

     # The first 5 elements
     print(array[:5])

     # The last 5 elements
     print(array[-5:])

     # Elements at even indices
     print(array[1::2])

[1 2 3 4 5]
[16 17 18 19 20]
[ 2  4  6  8 10 12 14 16 18 20]
```

#Answer 7

a = np.array([1, 2, 3])
b = np.array([4, 5, 6])

print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)

```
[12] a = np.array([1, 2, 3])
     b = np.array([4, 5, 6])

     print("Addition:", a + b)
     print("Subtraction:", a - b)
     print("Multiplication:", a * b)
     print("Division:", a / b)
```

```
Addition: [5 7 9]
Subtraction: [-3 -3 -3]
Multiplication: [ 4 10 18]
Division: [0.25 0.4  0.5 ]
```

#Answer 8

data = np.array([10, 15, 7, 22, 17])

# Mean
print("Mean:", np.mean(data))

# Standard deviation
print("Standard Deviation:", np.std(data))

# Median
print("Median:", np.median(data))

# Sum of all elements
print("Sum:", np.sum(data))

```
data = np.array([10, 15, 7, 22, 17])

# Mean
print("Mean:", np.mean(data))

# Standard deviation
print("Standard Deviation:", np.std(data))

# Median
print("Median:", np.median(data))

# Sum of all elements
print("Sum:", np.sum(data))
```

```
Mean: 14.2
Standard Deviation: 5.268775948927796
Median: 15.0
Sum: 71
```